

Juan Roberto García Gómez

Data Engineer | MES | Industrial IoT | Manufacturing Analytics

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PROFESSIONAL SUMMARY

Data Engineer specializing in MES, Industrial IoT, and manufacturing analytics, with experience building production-ready systems that connect shopfloor execution, structured data, and management decision-making. Hands-on delivery experience spans PostgreSQL, SQL, ETL, Node.js, React, Power BI, and operational dashboards for manufacturing environments. Combines technical implementation, manufacturing logic, analytics, and business communication to support data-driven plant operations and management visibility.

SELECTED IMPACT

- Built and deployed a web-based MES used daily across plant operations, supporting Injection, Assembly, EPC, and execution workflows.
- Integrated 34 machines into real-time dashboards, historical analysis, and operational reporting workflows.
- Recovered MX\$15.2M in logistics claims within one year through structured analysis and evidence management.
- Delivered Power BI dashboards that reduced analysis time by 40% and improved management visibility.

PROFESSIONAL EXPERIENCE

Chunil Mexico

Oct 2024 - Present

Data Engineer | MES & Manufacturing Analytics

Responsible for building production-ready manufacturing systems and analytics workflows that support daily plant execution, visibility, and management follow-up.

- Designed and deployed a web-based MES that replaced spreadsheet-based production control with centralized workflows across Injection, Assembly, EPC, Tooling, Quality, Maintenance, and plant execution.
- Built PostgreSQL data models and Node.js services for machines, parts, molds, shifts, cycle times, production targets, downtime, quality, and reporting.
- Integrated 34 injection machines into real-time monitoring, historical analysis, and automated reporting workflows.
- Developed React applications and Power BI dashboards for OEE, plan adherence, downtime, quality, production control, and management visibility.
- Supported Industrial IoT integration using machine data to improve visibility into running status, cycle counts, production performance, and downtime.
- Automated shift-level reporting and management follow-up to reduce manual reporting effort and improve operational response.
- Partnered with operations, quality, maintenance, and leadership teams to translate manufacturing requirements into production-ready systems.
- Helped establish a data-driven operating model across plant execution, reporting, and continuous improvement.

Ramírez Valle & Asociados | General Motors SLP Account

Oct 2023 - 2024

Claims, Loss & Recovery Coordinator

Role focused on operational analytics, financial recovery, evidence control, supplier follow-up, and management reporting.

- Recovered MX\$15.2M in logistics claims within one year through structured case analysis and evidence management.
- Managed 100+ investigations across claims, supplier follow-up, evidence control, and executive reporting.
- Built Power BI dashboards that reduced analysis time by 40% and improved visibility into recovery status.
- Standardized documentation, tracking, and review routines to improve resolution time and reporting consistency.
- Connected operational evidence, financial impact, and management reporting to support decision-making.

CORE SKILLS

Data Engineering: PostgreSQL, SQL, ETL, data modeling, reporting logic, data quality.

Software Development: Node.js, Express, React, Vite, REST APIs, and operational web applications.

Manufacturing Systems: MES, Industrial IoT, OEE, downtime, plan adherence, cycle time analysis, and quality reporting.

Analytics: Power BI, KPI design, dashboard development, operational performance analysis.

Business / Operations: stakeholder communication, requirements translation, process standardization, and executive reporting.

EDUCATION & PROFESSIONAL TRAINING

- Full Stack Web Development Bootcamp, KeepCoding | Nov 2025 - Present
- Business Analytics, Tecnológico de Monterrey | Jan 2025 - Apr 2025
- Big Data, Artificial Intelligence & Machine Learning Bootcamp, KeepCoding | Jun 2023 - Jun 2024
- Diploma in Property and Casualty Claims, Instituto Mexicano Educativo de Seguros y Fianzas | Feb 2024 - Nov 2024
- Python for Data Science, IICO | May 2022 - Aug 2022
- Bachelor's Degree in Economics, Universidad Autónoma de San Luis Potosí | Aug 2018 - Jul 2023
- Bachelor's Degree in Industrial & Logistics Engineering, Universidad Interamericana para el Desarrollo | Aug 2014 - Jul 2017

LANGUAGES

Spanish: Native

English: Professional working proficiency